

Learning for Data Science

Course 3: Bias, Variance, Over and Underfitting

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September 17, 2026



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de Montréal



Mila

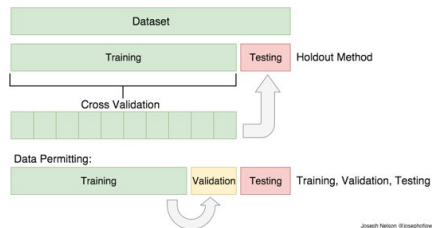
CIFAR

Outline

Over and under-fitting

What does it mean for a model to perform well?

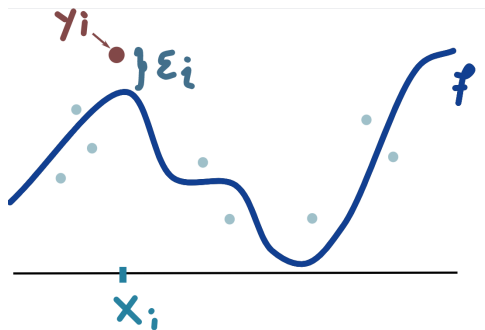
- Performance:
 - If we train a model to predict $p(\text{cat or dog} \mid x, \theta)$
- It should model this probability well
- We can evaluate this via the validation set
- General concept is Generalization
 - Discover the proper best factors for determining if an image contains a cat/dog
 - Good examples: Has whiskers, vertical iris, claws or not
 - Bad example: pixel 20,47 is gray
- We study this via a model's Bias and Variance



- Cross validation example

Learning Input - Output mapping

- $\hat{y}_i \leftarrow f(\mathbf{x}_i \mid \theta)$ is a “simple” function describing the relationship between x and y
- ϵ_i reflects variations whose source is unknown to us, because of not enough features or too complex a function — example: a coin toss



Bias

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Bias

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- We have collections of noisy data of a signal $f(x_i)$.

Bias

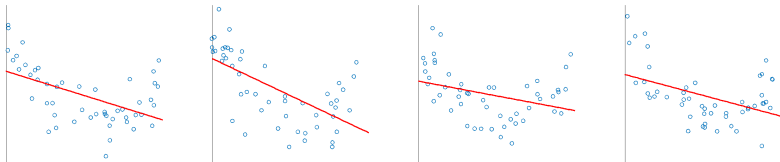
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- Regardless of the collection; the $f(x_i | \hat{\theta})$ model will produce systematic errors.

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- There are few difference between each model $\rightarrow$ low variance



Variance

- We try to train a more complex model $f(x_i | \hat{\theta})$.

Variance

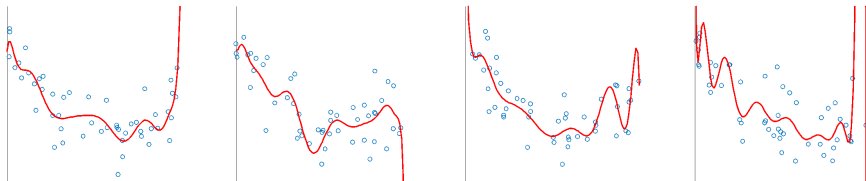
- We try to train a more complex model $f(x_i | \hat{\theta})$.
- On the noisy data from a signal $f(x_i)$.

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Variance

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- On the noisy data from a signal $f(x_i)$.
- Different datasets produce different model fits with unrealistically large oscillations.
- There are many variations between each model $\rightarrow$ high variance.



The Bias-Variance Trade-Off

Suppose we use M different collections of N data each:

$$\{\mathbf{x}_i^j, y_i^j\}, \quad j = 1, \dots, M$$

generated from a noisy function

$$y_i^j = f(\mathbf{x}_i^j) + e_i^j, \quad i = 1, \dots, N,$$

where $f(\mathbf{x})$ is an unknown function and e is measurement noise.

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where $f(\mathbf{x})$ is an unknown function and e is measurement noise.

- The values of $\mathbf{x}_i$ are the same for each game (just subsets).
- Let's separate the data into M training sets and M validation sets.

The Bias-Variance Trade-Off

For each set of N data we train a model $f(\mathbf{x} \mid \hat{\theta}_j)$.

The average of the models is

$$m(\mathbf{x}) = \frac{1}{M} \sum_{j=1}^M f(\mathbf{x} \mid \hat{\theta}_j)$$

The bias is calculated using the function:

$$b(\mathbf{x}) = m(\mathbf{x}) - f(\mathbf{x})$$

on the validation data. Ideally it should be zero everywhere.

The Bias-Variance Trade-off

The following measures are calculated with the bias and the validation data.

- $\text{Bias}^2(f(\mathbf{x} \mid \hat{\theta}_j))$:

$$\frac{1}{N} \sum_{i=1}^N [b(\mathbf{x}_i)]^2$$

- $\text{Variance}(f(\mathbf{x} \mid \hat{\theta}_j))$:

$$\frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M [m(\mathbf{x}_i) - f(\mathbf{x}_i \mid \hat{\theta}_j)]^2$$

- $\text{MSE}(f(\mathbf{x} \mid \hat{\theta}_j))$:

$$\frac{1}{MN} \sum_{i=1}^N \sum_{j=1}^M [y_i - f(\mathbf{x}_i \mid \hat{\theta}_j)]^2$$

The Bias-Variance Trade-off

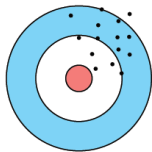
The following relationship between MSE (Mean Least Squares) and the other measures can be shown:

$$\text{MSE}(f(\mathbf{x} \mid \hat{\theta}_j)) = \text{Bias}^2(f(\mathbf{x} \mid \hat{\theta}_j)) + \text{Variance}(f(\mathbf{x} \mid \hat{\theta}_j)) + \sigma^2$$

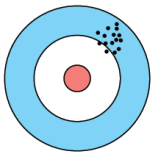
With σ^2 being the noise variance.

Further reading [here](#).

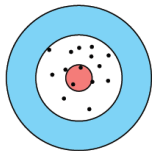
Examples of Distributions with Bias and Variance



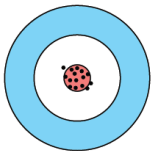
Biais et variance élevés



Biais élevé, variance faible



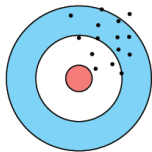
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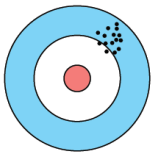
Biais et variance faibles

- Worst case scenario
- Performance of a player at the beginning of training

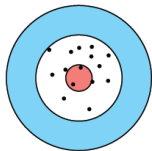
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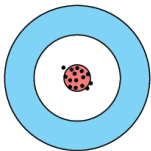
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Biais faible, variance élevée

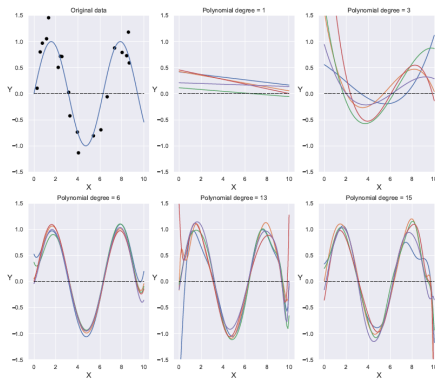


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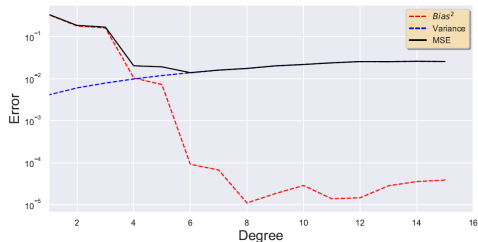
- Worst case scenario
- Performance of a player at the beginning of training
- Best scenario
- Performance of a well-trained player.
- Low bias and variance

Examples of polynomials of different degree

- low degree polynomials:
 - large bias
 - low variance
 - models are very different from the signal
- Large degree polynomials:
 - small bias
 - large variance (close to $x_i = 0$ and $x_i = 10$)
 - models interpolate the data



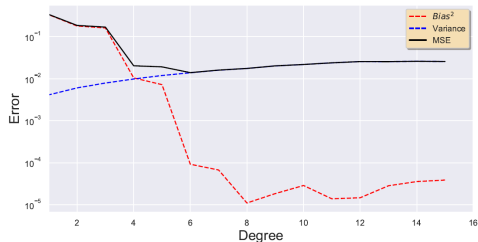
Examples of polynomials of different degree



- This figure shows how the mean squared error varies with the degree of the approximation polynomial for $f(x | \hat{\theta})$.
- The MSE is dominated by the bias of the **overly simple** model or the variance of the **overly complex model**.
- We use this curve to find the best model to analyze our data.

$$\hat{y}_i \leftarrow f(\mathbf{x}_i | \theta)$$

Examples of polynomials of different degree



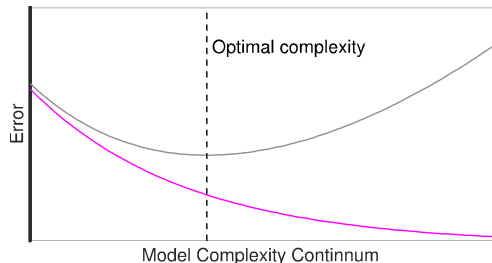
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$$\hat{y}_i \leftarrow f(\mathbf{x}_i | \theta)$$

- What is the best degree here?

Error Based on the Bias-Variance Trade-off

- A different approximate model $h_i(x_i)$ could have been used, but the results would have been similar.
- The following figure shows the general behaviour.
- The bias-variance trade-off is revealed via a validation set, not a train set!
 - $MSE_{validation}$, MSE_{train}



Bias and Variance Problems

- **Bias**

- Non-zero systematic error
- Overly simple models
- Reveals underfitting
- Measure: average of a model's predictions

Bias and Variance Problems

- **Bias**

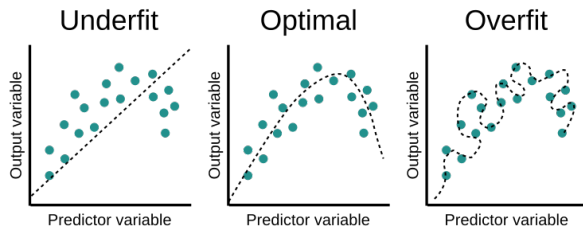
- Non-zero systematic error
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- **Variance**

- Good results on average but vary too much in practice
- Overly complex models
- Reveals overfitting
- Measure: average of a model's predictions

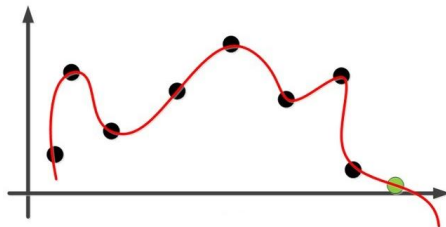
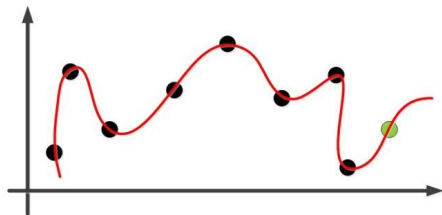
Over and under-fitting

- A model :
 - Learns details and noise in training data.
 - Produces good results on training data, but poor results on validation and test data.
 - Memorizes the data instead of learning and understanding the underlying trend of the data.
 - Is unable to generalize with new data.
 - Low bias but high variance.



The variance problem

- Overfitting is sometimes called the variance problem.
- A small change in the data (green dot) completely affects the model below.



Causes

- The model is too complex.



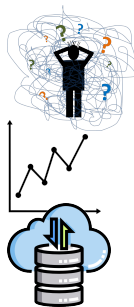
Causes

- The model is too complex.
- The model has a high variance.



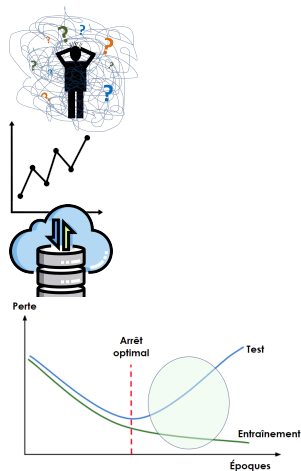
Causes

- The model is too complex.
- The model has a high variance.
- The size of the training dataset is not large enough.



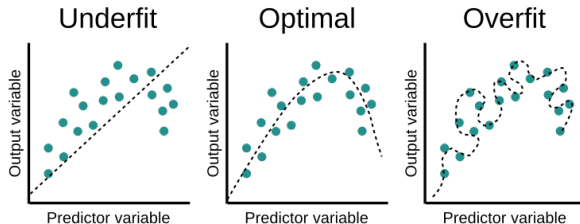
Causes

- The model is too complex.
- The model has a high variance.
- The size of the training dataset is not large enough.
- Trained too long!



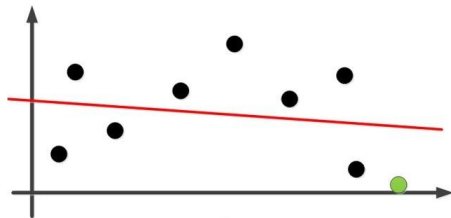
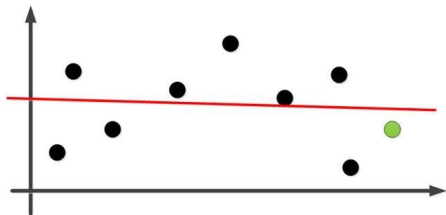
What is underfitting?

- Model too simple
 - Doesn't perform well with training and validation data.
 - Cannot grasp the relationship between input examples and target values.
- Low variance but high bias
- The model consistently makes similar errors in the same places.



Low variance, but high bias

- The pattern below varies little when moving the green dot.
- The model consistently makes similar errors in different places; it is therefore biased.



Causes

- The model is too simple.
- The model has a high bias.



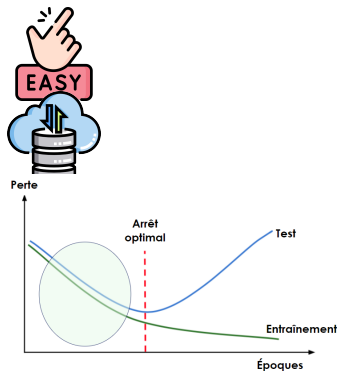
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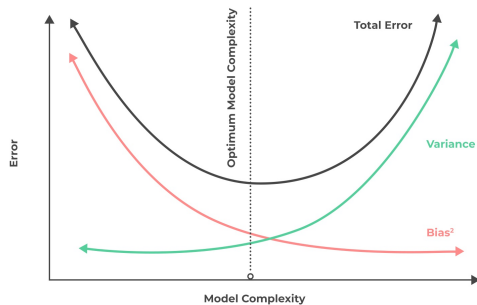


Causes

- The model is too simple.
- The model has a high bias.
- The size of the training dataset is not large enough.
- Not trained long enough!



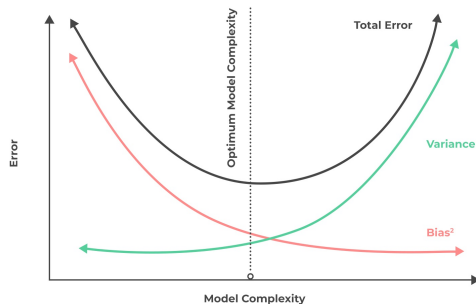
Bias-variance trade-off



- Under learning

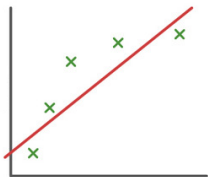
- Over learning

Bias-variance trade-off



- Under learning
 - An optimal point of model complexity exists where the bias and variance error curves intersect, and the error minimizes.
- Over learning

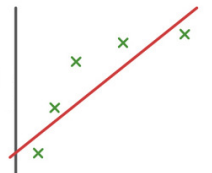
Examples in regression



$$\hat{y}_i = \beta_0 + \beta_1 x_i$$

- Under learning; model too simple to reproduce the data distribution.

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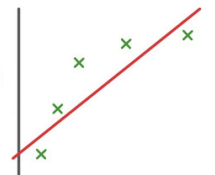
- Under learning; model too simple to reproduce the data distribution.



$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$$

- Good learning: good bias-variance compromise

Examples in regression



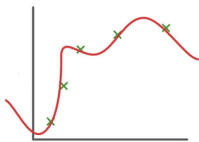
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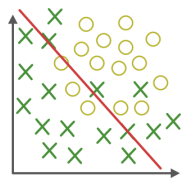
- Good learning: good bias-variance compromise



$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \beta_4 x_i^4$$

- Over learning; model so flexible that it reproduces the distribution of noisy data.

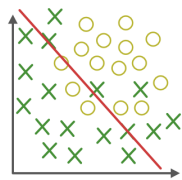
Examples in classification



Under-fitting

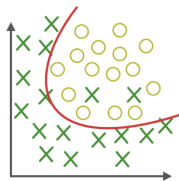
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Examples in classification



Under-fitting

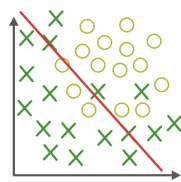
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Appropriate-fitting

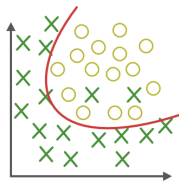
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Examples in classification



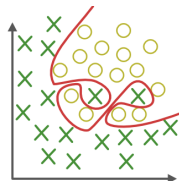
Under-fitting

- Under learning; model too simple to reproduce the data distribution.



Appropriate-fitting

- Good learning: good bias-variance compromise



Over-fitting

- Over learning; model so flexible that it reproduces the distribution of noisy data.

Solutions for under-fitting

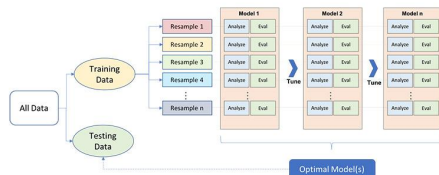
- Add features
- Add feature interactions
- Use a more complex
- Change model

Solutions for over-fitting

- Remove features
- Dimensionality reduction
- Regularization
- Add data
- Clean data (outliers)
- Simplify the model
- Change model type

Useful strategy to find the model with optimal complexity

- Several models are trained with the same training data.
- Their performance in generalization is measured with the same validation data.
- The model with the best performance is the one with the optimal complexity.

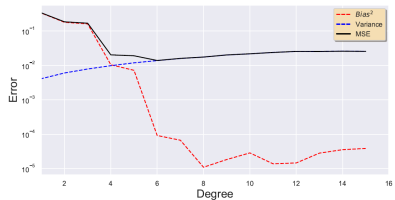


In-class poll: Bias-Variance Trade-off

- Log in with your first name, and your student number as your last name.

Q1: Best polynomial degree for this model?

(A) 2 (B) 4 (C) 6 (D) 10



In-class poll: Bias-Variance Trade-off

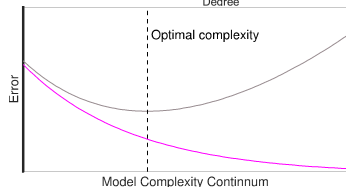
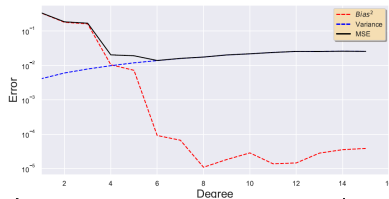
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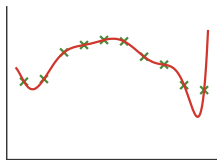
Q2: Labels removed — which curve is the validation set?

(A) gray (B) magenta (C) both (D) cannot tell



In-class poll: Which fit is which?

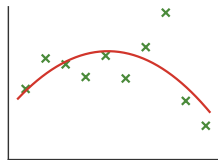
(A)



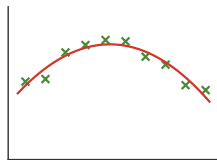
(B)



(C)



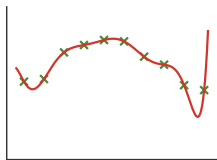
(D)



- **Q1:** Which plot shows underfitting?

In-class poll: Which fit is which?

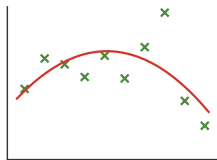
(A)



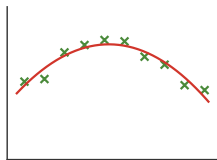
(B)



(C)



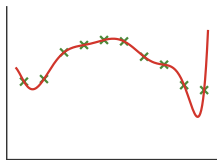
(D)



- **Q1:** Which plot shows underfitting?
- **Q2:** Which plot shows overfitting?

In-class poll: Which fit is which?

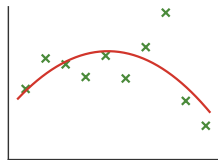
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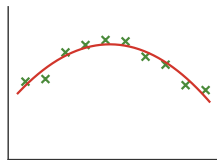
(B)



(C)



(D)



- **Q1:** Which plot shows underfitting?
- **Q2:** Which plot shows overfitting?
- **Q3:** Which plot is a good fit, despite noisy data?

Sources

- IVADO MOOC